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Wright et al.

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(54) **TRAINING AN ARTIFICIAL INTELLIGENCE ENGINE FOR REAL-TIME MONITORING TO ELIMINATE FALSE POSITIVES**

(58) **Field of Classification Search**

CPC .. G06F 21/50; G06F 2221/2117; G06N 3/047

(Continued)

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(56)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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G06N 3/045 (2023.01)

(Continued)

(52) **U.S. Cl.**

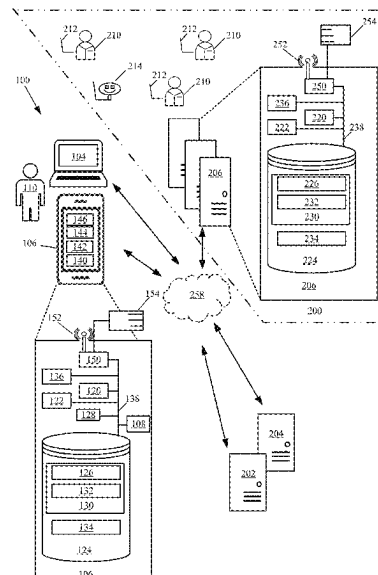
CPC **G06F 21/50** (2013.01); **G06N 3/047** (2023.01); **G06F 2221/2117** (2013.01)

(57)

ABSTRACT

A system and method for training an artificial intelligence engine for real-time monitoring to eliminate false positives is disclosed. The system includes at least one processor, a communication interface coupled to the processor, and a memory device storing executable code. Executing the executable code causes the processor to receive data from an AI security model, receive data from a false positive database, and correlate both sets of data. The correlated data is used to generate a training dataset and a test dataset used to train a false positive identification model. After evaluating the false positive identification model, an AI engine is applied to user registration. The AI engine includes an AI security model and the false positive identification model. Additionally, a system for evaluating the security of user registration utilizing the false positive identification model is disclosed.

18 Claims, 8 Drawing Sheets



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(58) **Field of Classification Search**

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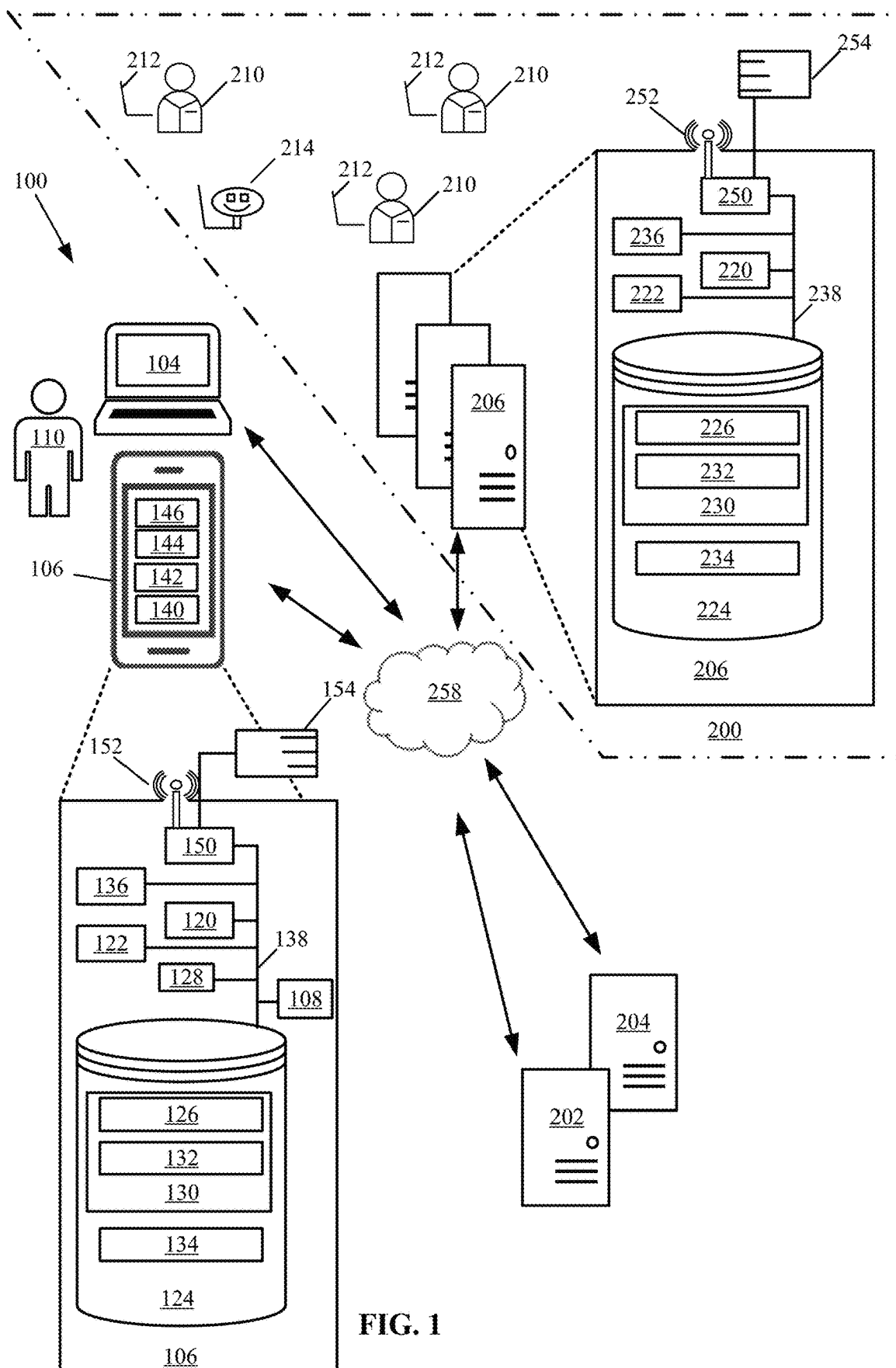
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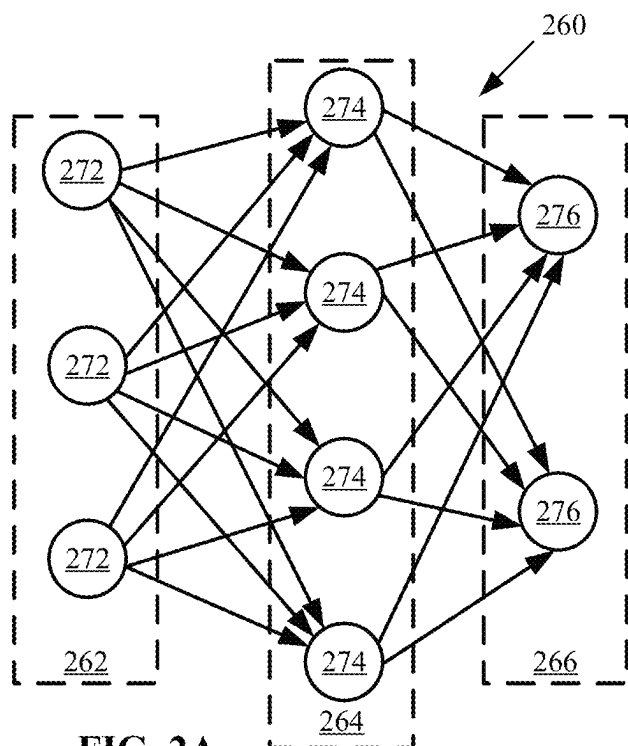


FIG. 2A

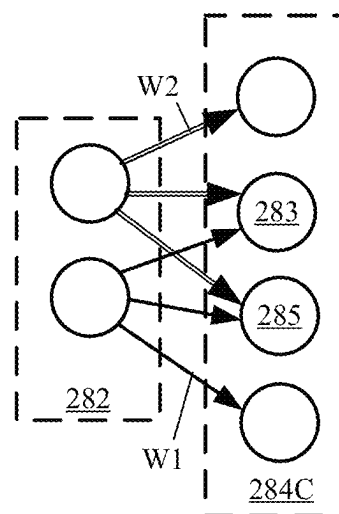


FIG. 2C

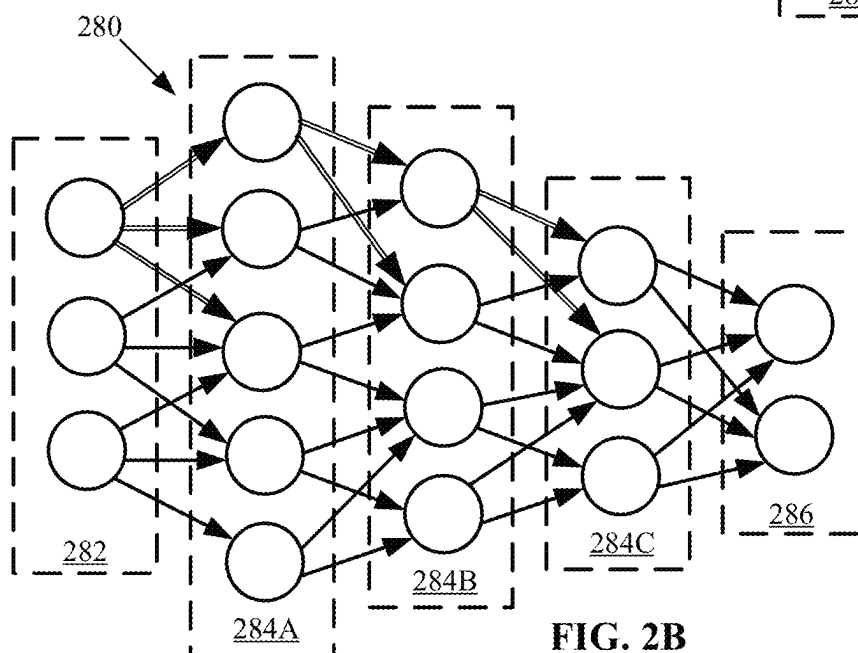


FIG. 2B

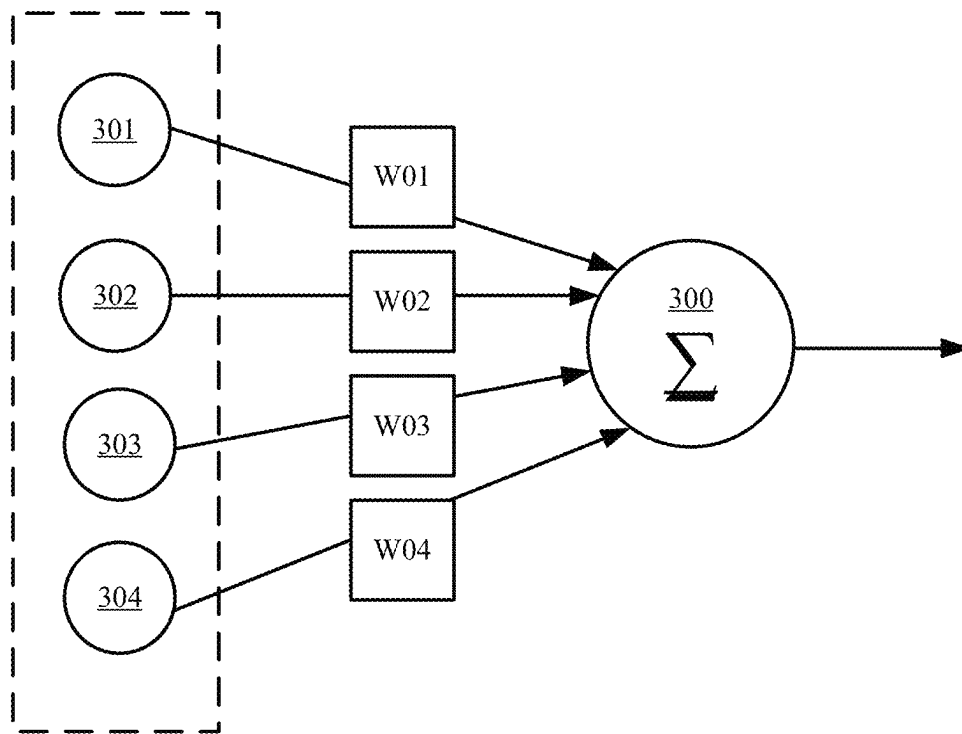


FIG. 3

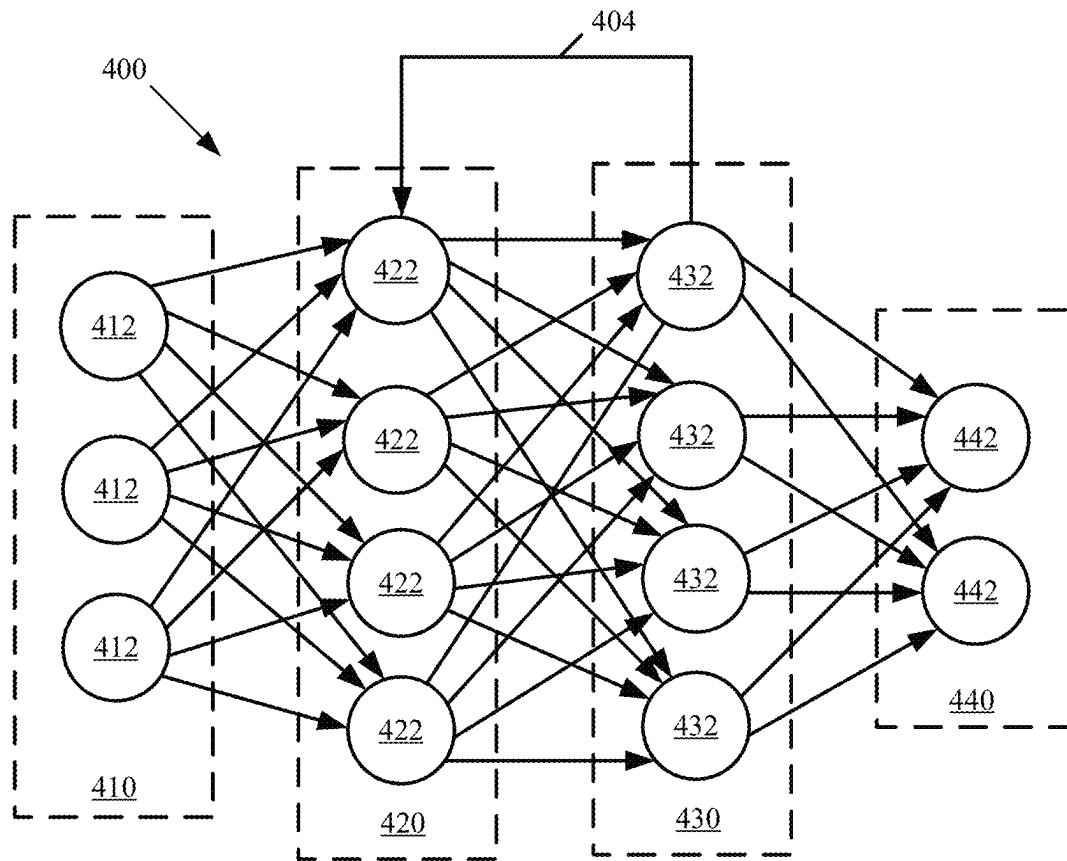


FIG. 4

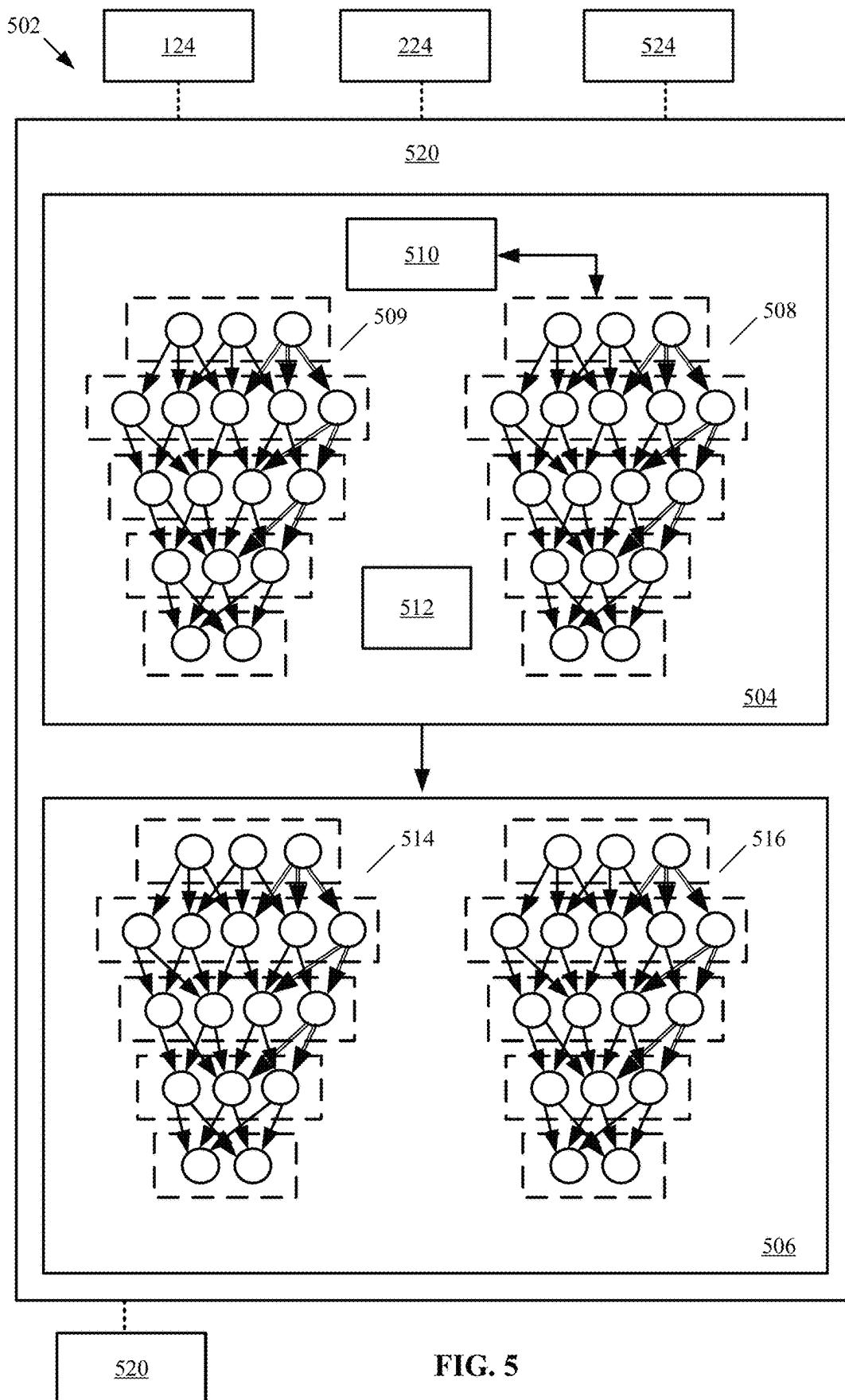


FIG. 5

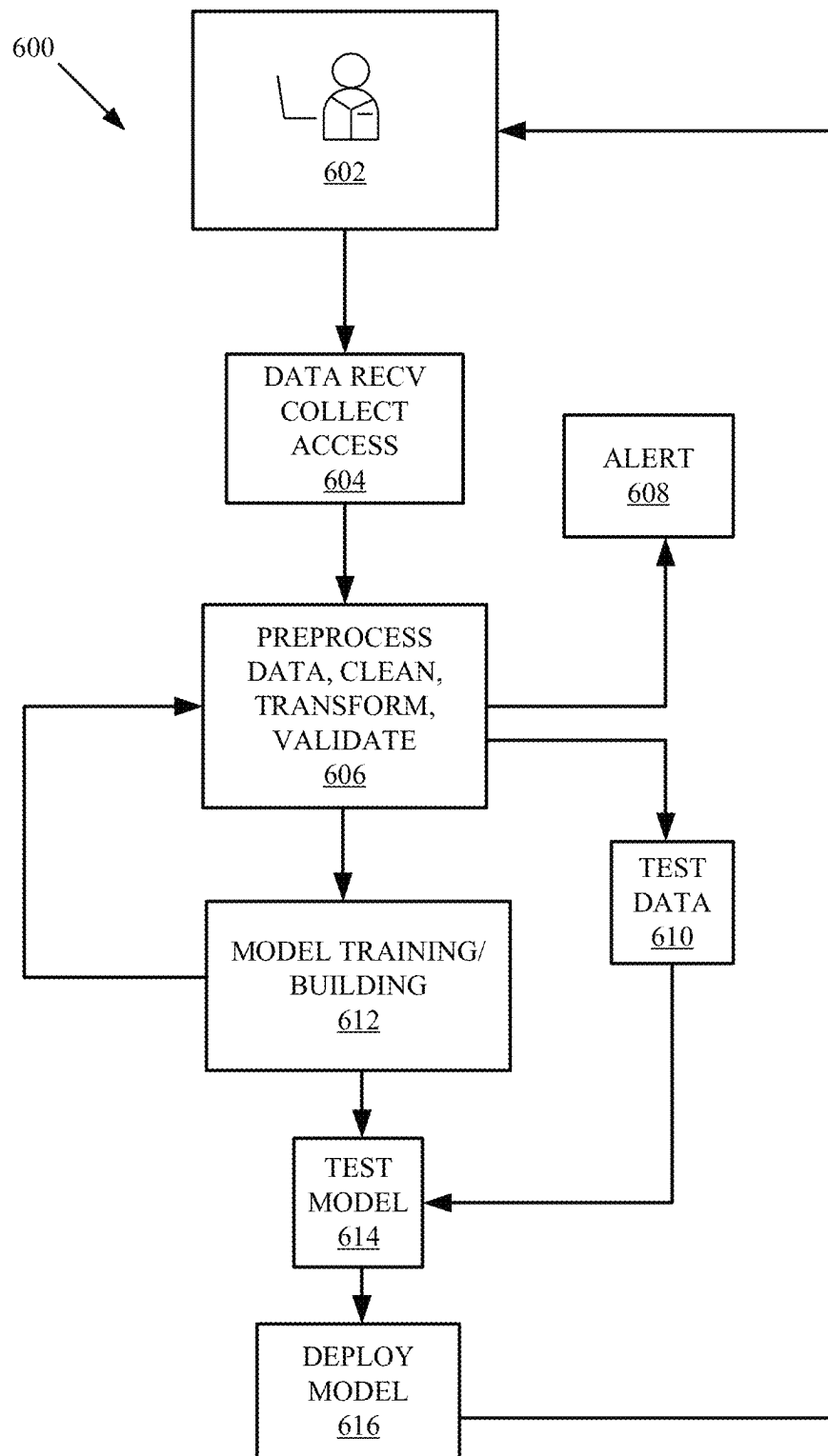


FIG. 6

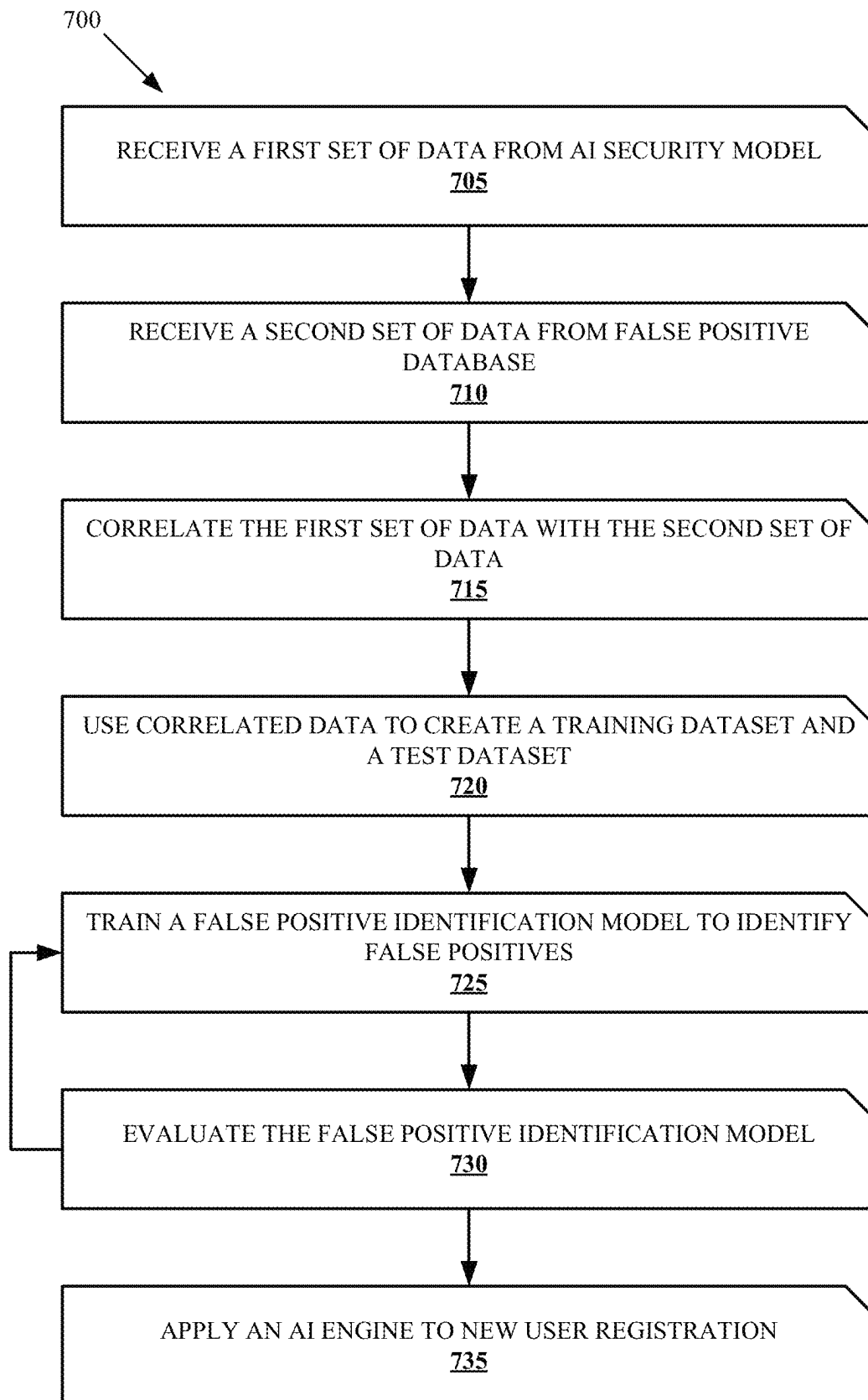


FIG. 7

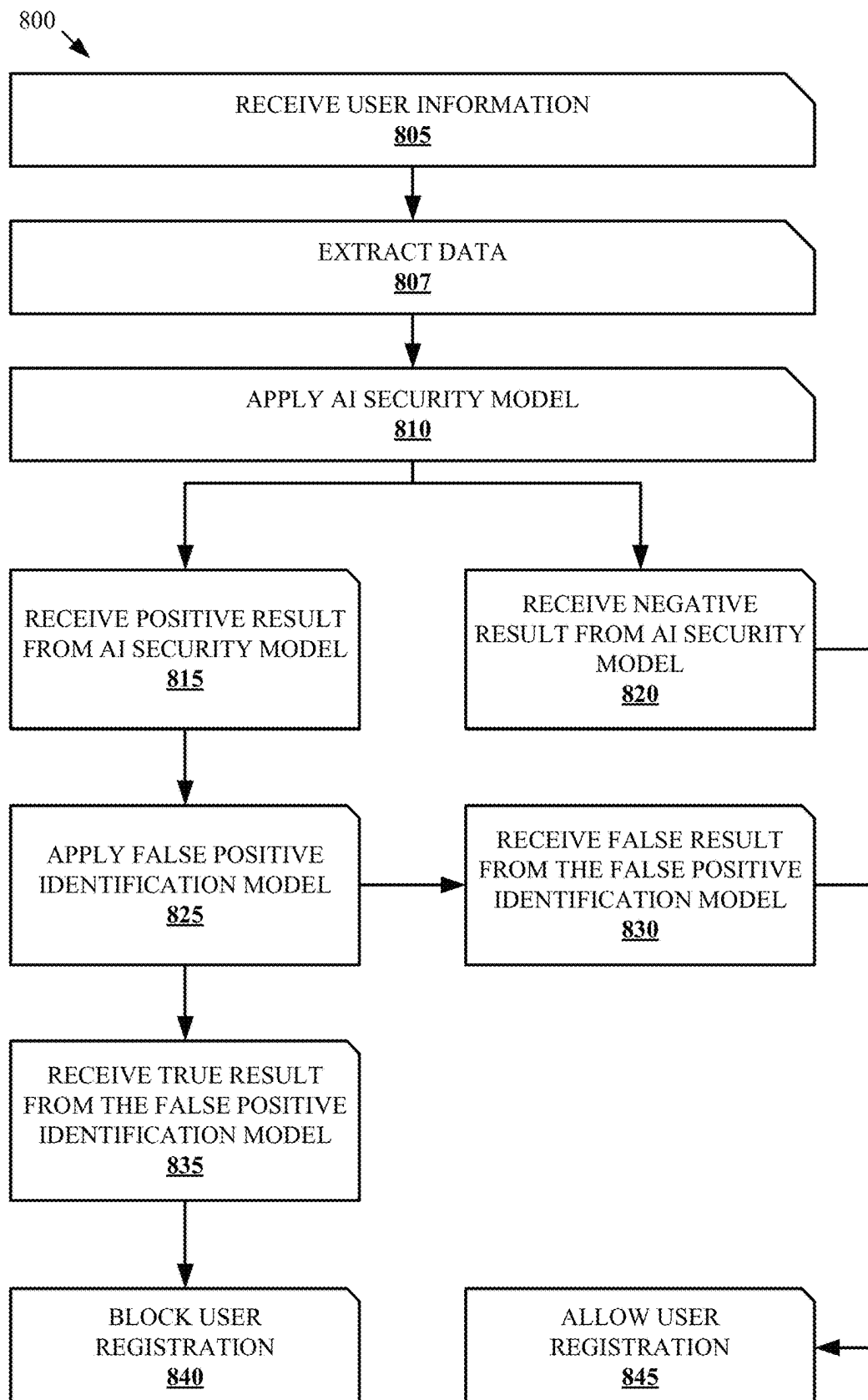


FIG. 8

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TRAINING AN ARTIFICIAL INTELLIGENCE ENGINE FOR REAL-TIME MONITORING TO ELIMINATE FALSE POSITIVES

FIELD

This invention relates generally to the field reducing false positives, and more particularly embodiments of the invention relate to reducing false positives from a security focused AI model.

BACKGROUND

Many organizations utilize machine learning, or artificial intelligence, methods for monitoring security during the creation of user accounts. None of these methods are completely free of false positives. A false positive is defined as a result incorrectly identified as positive. Typically, when a user registration is identified as positive for fraudulent or insecure activity, the user is blocked from creating an account. The only recourse a user may have in the case of a false positive is to contact a person within the organization, either by phone or in person. This action takes time from not only the organization, but also the user. The current disclosure addresses the problem of false positives by detecting and addressing false positives in real time.

BRIEF SUMMARY

Embodiments of the present invention address the above needs and/or achieve other advantages by providing apparatuses and methods that identify and reduce false positives in security related AI models.

In a first aspect, the present invention is related to a system for training an AI engine for real-time monitoring of false positives during user registration. The system consists of at least one processor, a communication interface communicatively coupled to the at least one processor, and a memory device storing executable code causing the processor to: receive a first set of data from at least one AI security model and a second set of data from a false positive database. The data received from the AI security model includes: input data indicative of user registration, and output data—a classification for the user registration. The second set of data includes a list of false positive instances from a false positive database. The executable code further causes the processor to correlate the first and second sets of data and create a training dataset and a test dataset from the correlated data. The executable code further causes the processor to use the training dataset to train a false positive identification model to identify false positives, and use the test dataset to evaluate the false positive identification model. Based on the evaluation, the executable code may cause the processor to apply an AI engine to user registration, where the AI engine includes an AI security model and the false positive identification model.

In another aspect of the present invention a method training an AI engine for real-time monitoring of false positives during user registration is presented. The method includes: receiving a first set of data from at least one AI security model and a second set of data from a false positive database. The first set of data includes input and output data from the AI security model. The second set of data includes false positive instances reported by users. The method further includes correlating the first and second sets of data and generating a training dataset and a test dataset. The method then uses the training dataset to train a false positive

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identification model and uses the test dataset to evaluate the false positive identification model, prior to applying an AI engine to user registration, where the AI engine includes both an AI security model and a false positive identification model.

In some embodiments, applying the AI engine includes: receiving user information and extracting data from the user information; applying the at least one AI security model to the extracted data and receiving a result from the at least one AI security model. The result may be a negative or a positive. Upon receiving a negative result, user registration is allowed. Upon receiving a positive result, the false negative identification model is applied. The false negative identification model will provide a result, either a false or a true result. If a false result is obtained the user registration is allowed. If the true result is obtained, the user registration is blocked. In some embodiments, applying the AI engine further comprises determining a false positive rate and based on the false positive rate triggering an alert to modify the at least one AI security model. In some embodiments, modifying the AI security model may include retraining the AI security model. Additionally, or alternatively, modifying the AI security model may include selecting a new AI security model.

In yet another aspect, the present invention provides a system for eliminating false positives during user registration. The system consists of at least one processor, a communication interface communicatively coupled to the at least one processor, and a memory device storing executable code causing the processor to: receive user information from a user attempting to register a new account, and apply an AI security model to the user information. The AI security model is configured to determine whether or not the user is attempting fraud. The executable code further causes the processor to receive a result from the at least one AI security model, where the result is either positive or negative. If the result is positive, a false positive identification model will be applied. A result will be received from the false positive identification model; if the result is false, the processor will allow the user registration. If the result is true the user registration will be blocked.

According to embodiments of the invention, the false positive identification model comprises a neural network. In some embodiments, the neural network includes a convolutional neural network (CNN). Additionally, or alternatively, the neural network includes a recurrent neural network (RNN). In some embodiments, the neural network includes a feed-forward network. According to some embodiments of the invention, the false positive identification model contains a Bayesian machine learning algorithm.

In some embodiments, the at least one AI security model is configured to compare the data from the user information to personally identifiable information from one or more public or private databases.

The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present invention or may be combined in yet other embodiments, further details of which can be seen with reference to the following description and drawings.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings, wherein:

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FIG. 1 illustrates an enterprise system, and environment thereof, according to at least one embodiment.

FIG. 2A is a diagram of a feedforward network, according to at least one embodiment, utilized in machine learning

FIG. 2B is a diagram of a convolution neural network, according to at least one embodiment, utilized in machine learning.

FIG. 2C is a diagram of a portion of the convolution neural network of FIG. 2B, according to at least one embodiment, illustrating assigned weights at connections or neurons.

FIG. 3 is a diagram representing an exemplary weighted sum computation in a node in an artificial neural network.

FIG. 4 is a diagram of a Recurrent Neural Network RNN, according to at least one embodiment, utilized in machine learning.

FIG. 5 is a schematic logic diagram of an artificial intelligence program including a front-end and a back-end algorithm.

FIG. 6 is a flow chart representing a method, according to at least one embodiment, of model development and deployment by machine learning.

FIG. 7 is a flow chart representing a method, according to at least one embodiment, of training an AI to detect false positives.

FIG. 8 is a flow chart representing a method, according to at least one embodiment, of applying an AI engine to eliminate false positives.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

Embodiments of the present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout. Unless described or implied as exclusive alternatives, features throughout the drawings and descriptions should be taken as cumulative, such that features expressly associated with some particular embodiments can be combined with other embodiments. Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which the presently disclosed subject matter pertains.

The exemplary embodiments are provided so that this disclosure will be both thorough and complete, and will fully convey the scope of the invention and enable one of ordinary skill in the art to make, use, and practice the invention.

The terms “coupled,” “fixed,” “attached to,” “communicatively coupled to,” “operatively coupled to,” and the like refer to both (i) direct connecting, coupling, fixing, attaching, communicatively coupling; and (ii) indirect connecting coupling, fixing, attaching, communicatively coupling via one or more intermediate components or features, unless otherwise specified herein. “Communicatively coupled to” and “operatively coupled to” can refer to physically and/or electrically related components.

Embodiments of the present invention described herein, with reference to flowchart illustrations and/or block diagrams of methods or apparatuses (the term “apparatus” includes systems and computer program products), will be understood such that each block of the flowchart illustrations

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and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a particular machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create mechanisms for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer readable memory produce an article of manufacture including instructions, which implement the function/act specified in the flowchart and/or block diagram block or blocks.

The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions, which execute on the computer or other programmable apparatus, provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. Alternatively, computer program implemented steps or acts may be combined with operator or human implemented steps or acts in order to carry out an embodiment of the invention.

While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations, modifications, and combinations of the herein described embodiments can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the included claims, the invention may be practiced other than as specifically described herein.

The present disclosure identifies and reduces the occurrence of false positives during the security processes that occur when a user registers for an account. Security processes are in place to prevent fraudulent activity from occurring. Often, when a user registration activity is registered as positive for fraud, the user will simply give up and attempt to register with another entity. Thus, the entity loses potential users because of strict security processes. In an effort to retain these users, it is imperative to detect and prevent false positives so that these users will continue their registration with the entity.

FIG. 1 illustrates a system 100 and environment thereof, according to at least one embodiment, by which a user 110 benefits through use of services and products of an enterprise system 200. The user 110 accesses services and products by use of one or more user devices, illustrated in separate examples as a computing device 104 and a mobile device 106, which may be, as non-limiting examples, a smart phone, a portable digital assistant (PDA), a pager, a mobile television, a gaming device, a laptop computer, a camera, a video recorder, an audio/video player, radio, a

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GPS device, or any combination of the aforementioned, or other portable device with processing and communication capabilities. In the illustrated example, the mobile device **106** is illustrated in FIG. 1 as having exemplary elements, the below descriptions of which apply as well to the computing device **104**, which can be, as non-limiting examples, a desktop computer, a laptop computer, or other user-accessible computing device.

Furthermore, the user device, referring to either or both of the computing device **104** and the mobile device **106**, may be or include a workstation, a server, or any other suitable device, including a set of servers, a cloud-based application or system, or any other suitable system, adapted to execute, for example any suitable operating system, including Linux, UNIX, Windows, macOS, IOS, Android and any other known operating system used on personal computers, central computing systems, phones, and other devices.

The user **110** can be an individual, a group, or any entity in possession of or having access to the user device, referring to either or both of the mobile device **104** and computing device **106**, which may be personal or public items. Although the user **110** may be singly represented in some drawings, at least in some embodiments according to these descriptions the user **110** is one of many such that a market or community of users, consumers, customers, business entities, government entities, clubs, and groups of any size are all within the scope of these descriptions.

The user device, as illustrated with reference to the mobile device **106**, includes components such as, at least one of each of a processing device **120**, and a memory device **122** for processing use, such as random access memory (RAM), and read-only memory (ROM). The illustrated mobile device **106** further includes a storage device **124** including at least one of a non-transitory storage medium, such as a microdrive, for long-term, intermediate-term, and short-term storage of computer-readable instructions **126** for execution by the processing device **120**. For example, the instructions **126** can include instructions for an operating system and various applications or programs **130**, of which the application **132** is represented as a particular example. The storage device **124** can store various other data items **134**, which can include, as non-limiting examples, cached data, user files such as those for pictures, audio and/or video recordings, files downloaded or received from other devices, and other data items preferred by the user or required or related to any or all of the applications or programs **130**.

The memory device **122** is operatively coupled to the processing device **120**. As used herein, memory includes any computer readable medium to store data, code, or other information. The memory device **122** may include volatile memory, such as volatile Random Access Memory (RAM) including a cache area for the temporary storage of data. The memory device **122** may also include non-volatile memory, which can be embedded and/or may be removable. The non-volatile memory can additionally or alternatively include an electrically erasable programmable read-only memory (EEPROM), flash memory or the like.

The memory device **122** and storage device **124** can store any of a number of applications which comprise computer-executable instructions and code executed by the processing device **120** to implement the functions of the mobile device **106** described herein. For example, the memory device **122** may include such applications as a conventional web browser application and/or a mobile P2P payment system client application. These applications also typically provide a graphical user interface (GUI) on the display **140** that allows the user **110** to communicate with the mobile device

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106, and, for example a mobile banking system, and/or other devices or systems. In one embodiment, when the user **110** decides to enroll in a mobile banking program, the user **110** downloads or otherwise obtains the mobile banking system client application from a mobile banking system, for example enterprise system **200**, or from a distinct application server. In other embodiments, the user **110** interacts with a mobile banking system via a web browser application in addition to, or instead of, the mobile P2P payment system client application.

The processing device **120**, and other processors described herein, generally include circuitry for implementing communication and/or logic functions of the mobile device **106**. For example, the processing device **120** may include a digital signal processor, a microprocessor, and various analog to digital converters, digital to analog converters, and/or other support circuits. Control and signal processing functions of the mobile device **106** are allocated between these devices according to their respective capabilities. The processing device **120** thus may also include the functionality to encode and interleave messages and data prior to modulation and transmission. The processing device **120** can additionally include an internal data modem. Further, the processing device **120** may include functionality to operate one or more software programs, which may be stored in the memory device **122**, or in the storage device **124**. For example, the processing device **120** may be capable of operating a connectivity program, such as a web browser application. The web browser application may then allow the mobile device **106** to transmit and receive web content, such as, for example, location-based content and/or other web page content, according to a Wireless Application Protocol (WAP), Hypertext Transfer Protocol (HTTP), and/or the like.

The memory device **122** and storage device **124** can each also store any of a number of pieces of information, and data, used by the user device and the applications and devices that facilitate functions of the user device, or are in communication with the user device, to implement the functions described herein and others not expressly described. For example, the storage device may include such data as user authentication information, etc.

The processing device **120**, in various examples, can operatively perform calculations, can process instructions for execution, and can manipulate information. The processing device **120** can execute machine-executable instructions stored in the storage device **124** and/or memory device **122** to thereby perform methods and functions as described or implied herein, for example by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subject matters of these descriptions pertain. The processing device **120** can be or can include, as non-limiting examples, a central processing unit (CPU), a microprocessor, a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a programmable logic device (PLD), a digital signal processor (DSP), a field programmable gate array (FPGA), a state machine, a controller, gated or transistor logic, discrete physical hardware components, and combinations thereof. In some embodiments, particular portions or steps of methods and functions described herein are performed in whole or in part by way of the processing device **120**, while in other embodiments methods and functions described herein include cloud-based computing in whole or in part such that the processing device **120** facilitates local operations including, as non-limiting examples, communication, data transfer, and user

inputs and outputs such as receiving commands from and providing displays to the user.

The mobile device **106**, as illustrated, includes an input and output system **136**, referring to, including, or operatively coupled with, user input devices and user output devices, which are operatively coupled to the processing device **120**. The user output devices include a display **140** (e.g., a liquid crystal display or the like), which can be, as a non-limiting example, a touch screen of the mobile device **106**, which serves both as an output device, by providing graphical and text indicia and presentations for viewing by one or more user **110**, and as an input device, by providing virtual buttons, selectable options, a virtual keyboard, and other indicia that, when touched, control the mobile device **106** by user action. The user output devices include a speaker **144** or other audio device. The user input devices, which allow the mobile device **106** to receive data and actions such as button manipulations and touches from a user such as the user **110**, may include any of a number of devices allowing the mobile device **106** to receive data from a user, such as a keypad, keyboard, touch-screen, touchpad, microphone **142**, mouse, joystick, other pointer device, button, soft key, and/or other input device(s). The user interface may also include a camera **146**, such as a digital camera.

Further non-limiting examples include, one or more of each, any, and all of a wireless or wired keyboard, a mouse, a touchpad, a button, a switch, a light, an LED, a buzzer, a bell, a printer and/or other user input devices and output devices for use by or communication with the user **110** in accessing, using, and controlling, in whole or in part, the user device, referring to either or both of the computing device **104** and a mobile device **106**. Inputs by one or more user **110** can thus be made via voice, text or graphical indicia selections. For example, such inputs in some examples correspond to user-side actions and communications seeking services and products of the enterprise system **200**, and at least some outputs in such examples correspond to data representing enterprise-side actions and communications in two-way communications between a user **110** and an enterprise system **200**.

The mobile device **106** may also include a positioning device **108**, which can be for example a global positioning system device (GPS) configured to be used by a positioning system to determine a location of the mobile device **106**. For example, the positioning system device **108** may include a GPS transceiver. In some embodiments, the positioning system device **108** includes an antenna, transmitter, and receiver. For example, in one embodiment, triangulation of cellular signals may be used to identify the approximate location of the mobile device **106**. In other embodiments, the positioning device **108** includes a proximity sensor or transmitter, such as an RFID tag, that can sense or be sensed by devices known to be located proximate a merchant or other location to determine that the consumer mobile device **106** is located proximate these known devices.

In the illustrated example, a system intraconnect **138**, connects, for example electrically, the various described, illustrated, and implied components of the mobile device **106**. The intraconnect **138**, in various non-limiting examples, can include or represent, a system bus, a high-speed interface connecting the processing device **120** to the memory device **122**, individual electrical connections among the components, and electrical conductive traces on a motherboard common to some or all of the above-described components of the user device. As discussed herein, the system intraconnect **138** may operatively couple various

components with one another, or in other words, electrically connects those components, either directly or indirectly—with one another.

The user device, referring to either or both of the computing device **104** and the mobile device **106**, with particular reference to the mobile device **106** for illustration purposes, includes a communication interface **150**, by which the mobile device **106** communicates and conducts transactions with other devices and systems. The communication interface **150** may include digital signal processing circuitry and may provide two-way communications and data exchanges, for example wirelessly via wireless communication device **152**, and for an additional or alternative example, via wired or docked communication by mechanical electrically conductive connector **154**. Communications may be conducted via various modes or protocols, of which GSM voice calls, SMS, EMS, MMS messaging, TDMA, CDMA, PDC, WCDMA, CDMA2000, and GPRS, are all non-limiting and non-exclusive examples. Thus, communications can be conducted, for example, via the wireless communication device **152**, which can be or include a radio-frequency transceiver, a Bluetooth device, Wi-Fi device, a Near-field communication device, and other transceivers. In addition, GPS (Global Positioning System) may be included for navigation and location-related data exchanges, ingoing and/or outgoing. Communications may also or alternatively be conducted via the connector **154** for wired connections such by USB, Ethernet, and other physically connected modes of data transfer.

The processing device **120** is configured to use the communication interface **150** as, for example, a network interface to communicate with one or more other devices on a network. In this regard, the communication interface **150** utilizes the wireless communication device **152** as an antenna operatively coupled to a transmitter and a receiver (together a “transceiver”) included with the communication interface **150**. The processing device **120** is configured to provide signals to and receive signals from the transmitter and receiver, respectively. The signals may include signaling information in accordance with the air interface standard of the applicable cellular system of a wireless telephone network. In this regard, the mobile device **106** may be configured to operate with one or more air interface standards, communication protocols, modulation types, and access types. By way of illustration, the mobile device **106** may be configured to operate in accordance with any of a number of first, second, third, fourth, fifth-generation communication protocols and/or the like. For example, the mobile device **106** may be configured to operate in accordance with second-generation (2G) wireless communication protocols IS-136 (time division multiple access (TDMA)), GSM (global system for mobile communication), and/or IS-95 (code division multiple access (CDMA)), or with third-generation (3G) wireless communication protocols, such as Universal Mobile Telecommunications System (UMTS), CDMA2000, wideband CDMA (WCDMA) and/or time division-synchronous CDMA (TD-SCDMA), with fourth-generation (4G) wireless communication protocols such as Long-Term Evolution (LTE), fifth-generation (5G) wireless communication protocols, Bluetooth Low Energy (BLE) communication protocols such as Bluetooth 5.0, ultra-wideband (UWB) communication protocols, and/or the like. The mobile device **106** may also be configured to operate in accordance with non-cellular communication mechanisms, such as via a wireless local area network (WLAN) or other communication/data networks.

The communication interface **150** may also include a payment network interface. The payment network interface may include software, such as encryption software, and hardware, such as a modem, for communicating information to and/or from one or more devices on a network. For example, the mobile device **106** may be configured so that it can be used as a credit or debit card by, for example, wirelessly communicating account numbers or other authentication information to a terminal of the network. Such communication could be performed via transmission over a wireless communication protocol such as the Near-field communication protocol.

The mobile device **106** further includes a power source **128**, such as a battery, for powering various circuits and other devices that are used to operate the mobile device **106**. Embodiments of the mobile device **106** may also include a clock or other timer configured to determine and, in some cases, communicate actual or relative time to the processing device **120** or one or more other devices. For further example, the clock may facilitate timestamping transmissions, receptions, and other data for security, authentication, logging, polling, data expiry, and forensic purposes.

System **100** as illustrated diagrammatically represents at least one example of a possible implementation, where alternatives, additions, and modifications are possible for performing some or all of the described methods, operations and functions. Although shown separately, in some embodiments, two or more systems, servers, or illustrated components may be utilized. In some implementations, the functions of one or more systems, servers, or illustrated components may be provided by a single system or server. In some embodiments, the functions of one illustrated system or server may be provided by multiple systems, servers, or computing devices, including those physically located at a central facility, those logically local, and those located as remote with respect to each other.

The enterprise system **200** can offer any number or type of services and products to one or more users **110**. In some examples, an enterprise system **200** offers products. In some examples, an enterprise system **200** offers services. Use of “service(s)” or “product(s)” thus relates to either or both in these descriptions. With regard, for example, to online information and financial services, “service” and “product” are sometimes termed interchangeably. In non-limiting examples, services and products include retail services and products, information services and products, custom services and products, predefined or pre-offered services and products, consulting services and products, advising services and products, forecasting services and products, internet products and services, social media, and financial services and products, which may include, in non-limiting examples, services and products relating to banking, checking, savings, investments, credit cards, automatic-teller machines, debit cards, loans, mortgages, personal accounts, business accounts, account management, credit reporting, credit requests, and credit scores.

To provide access to, or information regarding, some or all the services and products of the enterprise system **200**, automated assistance may be provided by the enterprise system **200**. For example, automated access to user accounts and replies to inquiries may be provided by enterprise-side automated voice, text, and graphical display communications and interactions. In at least some examples, any number of human agents **210**, can be employed, utilized, authorized or referred by the enterprise system **200**. Such human agents **210** can be, as non-limiting examples, point of sale or point of service (POS) representatives, online cus-

tomers service assistants available to users **110**, advisors, managers, sales team members, and referral agents ready to route user requests and communications to preferred or particular other agents, human or virtual.

Human agents **210** may utilize agent devices **212** to serve users in their interactions to communicate and take action. The agent devices **212** can be, as non-limiting examples, computing devices, kiosks, terminals, smart devices such as phones, and devices and tools at customer service counters and windows at POS locations. In at least one example, the diagrammatic representation of the components of the user device **106** in FIG. 1 applies as well to one or both of the computing device **104** and the agent devices **212**.

Agent devices **212** individually or collectively include input devices and output devices, including, as non-limiting examples, a touch screen, which serves both as an output device by providing graphical and text indicia and presentations for viewing by one or more agent **210**, and as an input device by providing virtual buttons, selectable options, a virtual keyboard, and other indicia that, when touched or activated, control or prompt the agent device **212** by action of the attendant agent **210**. Further non-limiting examples include, one or more of each, any, and all of a keyboard, a mouse, a touchpad, a joystick, a button, a switch, a light, an LED, a microphone serving as input device for example for voice input by a human agent **210**, a speaker serving as an output device, a camera serving as an input device, a buzzer, a bell, a printer and/or other user input devices and output devices for use by or communication with a human agent **210** in accessing, using, and controlling, in whole or in part, the agent device **212**.

Inputs by one or more human agents **210** can thus be made via voice, text or graphical indicia selections. For example, some inputs received by an agent device **212** in some examples correspond to, control, or prompt enterprise-side actions and communications offering services and products of the enterprise system **200**, information thereof, or access thereto. At least some outputs by an agent device **212** in some examples correspond to, or are prompted by, user-side actions and communications in two-way communications between a user **110** and an enterprise-side human agent **210**.

From a user perspective experience, an interaction in some examples within the scope of these descriptions begins with direct or first access to one or more human agents **210** in person, by phone, or online for example via a chat session or website function or feature. In other examples, a user is first assisted by a virtual agent **214** of the enterprise system **200**, which may satisfy user requests or prompts by voice, text, or online functions, and may refer users to one or more human agents **210** once preliminary determinations or conditions are made or met.

A computing system **206** of the enterprise system **200** may include components such as, at least one of each of a processing device **220**, and a memory device **222** for processing use, such as random access memory (RAM), and read-only memory (ROM). The illustrated computing system **206** further includes a storage device **224** including at least one non-transitory storage medium, such as a micro-drive, for long-term, intermediate-term, and short-term storage of computer-readable instructions **226** for execution by the processing device **220**. For example, the instructions **226** can include instructions for an operating system and various applications or programs **230**, of which the application **232** is represented as a particular example. The storage device **224** can store various other data **234**, which can include, as non-limiting examples, cached data, and files such as those for user accounts, user profiles, account balances, and trans-

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action histories, files downloaded or received from other devices, and other data items preferred by the user or required or related to any or all of the applications or programs **230**.

The computing system **206**, in the illustrated example, includes an input/output system **236**, referring to, including, or operatively coupled with input devices and output devices such as, in a non-limiting example, agent devices **212**, which have both input and output capabilities.

In the illustrated example, a system intraconnect **238** electrically connects the various above-described components of the computing system **206**. In some cases, the intraconnect **238** operatively couples components to one another, which indicates that the components may be directly or indirectly connected, such as by way of one or more intermediate components. The intraconnect **238**, in various non-limiting examples, can include or represent, a system bus, a high-speed interface connecting the processing device **220** to the memory device **222**, individual electrical connections among the components, and electrical conductive traces on a motherboard common to some or all of the above-described components of the user device.

The computing system **206**, in the illustrated example, includes a communication interface **250**, by which the computing system **206** communicates and conducts transactions with other devices and systems. The communication interface **250** may include digital signal processing circuitry and may provide two-way communications and data exchanges, for example wirelessly via wireless device **252**, and for an additional or alternative example, via wired or docked communication by mechanical electrically conductive connector **254**. Communications may be conducted via various modes or protocols, of which GSM voice calls, SMS, EMS, MMS messaging, TDMA, CDMA, PDC, WCDMA, CDMA2000, and GPRS, are all non-limiting and non-exclusive examples. Thus, communications can be conducted, for example, via the wireless device **252**, which can be or include a radio-frequency transceiver, a Bluetooth device, Wi-Fi device, Near-field communication device, and other transceivers. In addition, GPS (Global Positioning System) may be included for navigation and location-related data exchanges, ingoing and/or outgoing. Communications may also or alternatively be conducted via the connector **254** for wired connections such as by USB, Ethernet, and other physically connected modes of data transfer.

The processing device **220**, in various examples, can operatively perform calculations, can process instructions for execution, and can manipulate information. The processing device **220** can execute machine-executable instructions stored in the storage device **224** and/or memory device **222** to thereby perform methods and functions as described or implied herein, for example by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subjects matters of these descriptions pertain. The processing device **220** can be or can include, as non-limiting examples, a central processing unit (CPU), a microprocessor, a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a programmable logic device (PLD), a digital signal processor (DSP), a field programmable gate array (FPGA), a state machine, a controller, gated or transistor logic, discrete physical hardware components, and combinations thereof.

Furthermore, the computing device **206**, may be or include a workstation, a server, or any other suitable device, including a set of servers, a cloud-based application or system, or any other suitable system, adapted to execute, for

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example any suitable operating system, including Linux, UNIX, Windows, macOS, IOS, Android, and any known other operating system used on personal computer, central computing systems, phones, and other devices.

The user devices, referring to either or both of the mobile device **104** and computing device **106**, the agent devices **212**, and the enterprise computing system **206**, which may be one or any number centrally located or distributed, are in communication through one or more networks, referenced as network **258** in FIG. 1.

Network **258** provides wireless or wired communications among the components of the system **100** and the environment thereof, including other devices local or remote to those illustrated, such as additional mobile devices, servers, and other devices communicatively coupled to network **258**, including those not illustrated in FIG. 1. The network **258** is singly depicted for illustrative convenience, but may include more than one network without departing from the scope of these descriptions. In some embodiments, the network **258** may be or provide one or more cloud-based services or operations. The network **258** may be or include an enterprise or secured network, or may be implemented, at least in part, through one or more connections to the Internet. A portion of the network **258** may be a virtual private network (VPN) or an Intranet. The network **258** can include wired and wireless links, including, as non-limiting examples, 802.11a/b/g/n/ac, 802.20, WiMax, LTE, and/or any other wireless link. The network **258** may include any internal or external network, networks, sub-network, and combinations of such operable to implement communications between various computing components within and beyond the illustrated environment **100**. The network **258** may communicate, for example, Internet Protocol (IP) packets, Frame Relay frames, Asynchronous Transfer Mode (ATM) cells, voice, video, data, and other suitable information between network addresses. The network **258** may also include one or more local area networks (LANs), radio access networks (RANs), metropolitan area networks (MANs), wide area networks (WANs), all or a portion of the internet and/or any other communication system or systems at one or more locations.

Two external systems **202** and **204** are expressly illustrated in FIG. 1, representing any number and variety of data sources, users, consumers, customers, business entities, banking systems, government entities, clubs, and groups of any size are all within the scope of the descriptions. In at least one example, the external systems **202** and **204** represent automatic teller machines (ATMs) utilized by the enterprise system **200** in serving users **110**. In another example, the external systems **202** and **204** represent payment clearinghouse or payment rail systems for processing payment transactions, and in another example, the external systems **202** and **204** represent third party systems such as merchant systems configured to interact with the user device **106** during transactions and also configured to interact with the enterprise system **200** in back-end transactions clearing processes.

In certain embodiments, one or more of the systems such as the user device **106**, the enterprise system **200**, and/or the external systems **202** and **204** are, include, or utilize virtual resources. In some cases, such virtual resources are considered cloud resources or virtual machines. Such virtual resources may be available for shared use among multiple distinct resource consumers and in certain implementations, virtual resources do not necessarily correspond to one or more specific pieces of hardware, but rather to a collection

of pieces of hardware operatively coupled within a cloud computing configuration so that the resources may be shared as needed.

As used herein, an artificial intelligence system, artificial intelligence algorithm, artificial intelligence module, program, and the like, generally refer to computer implemented programs that are suitable to simulate intelligent behavior (i.e., intelligent human behavior) and/or computer systems and associated programs suitable to perform tasks that typically require a human to perform, such as tasks requiring visual perception, speech recognition, decision-making, translation, and the like. An artificial intelligence system may include, for example, at least one of a series of associated if-then logic statements, a statistical model suitable to map raw sensory data into symbolic categories and the like, or a machine learning program. A machine learning program, machine learning algorithm, or machine learning module, as used herein, is generally a type of artificial intelligence including one or more algorithms that can learn and/or adjust parameters based on input data provided to the algorithm. In some instances, machine learning programs, algorithms, and modules are used at least in part in implementing artificial intelligence (AI) functions, systems, and methods.

Artificial Intelligence and/or machine learning programs may be associated with or conducted by one or more processors, memory devices, and/or storage devices of a computing system or device. It should be appreciated that the AI algorithm or program may be incorporated within the existing system architecture or be configured as a standalone modular component, controller, or the like communicatively coupled to the system. An AI program and/or machine learning program may generally be configured to perform methods and functions as described or implied herein, for example by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subjects matters of these descriptions pertain.

A machine learning program may be configured to implement stored processing, such as decision tree learning, association rule learning, artificial neural networks, recurrent artificial neural networks, long short term memory networks, inductive logic programming, support vector machines, clustering, Bayesian networks, reinforcement learning, representation learning, similarity and metric learning, sparse dictionary learning, genetic algorithms, k-nearest neighbor (KNN), and the like. In some embodiments, the machine learning algorithm may include one or more image recognition algorithms suitable to determine one or more categories to which an input, such as data communicated from a visual sensor or a file in JPEG, PNG or other format, representing an image or portion thereof, belongs. Additionally or alternatively, the machine learning algorithm may include one or more regression algorithms configured to output a numerical value given an input. Further, the machine learning may include one or more pattern recognition algorithms, e.g., a module, subroutine or the like capable of translating text or string characters and/or a speech recognition module or subroutine. In various embodiments, the machine learning module may include a machine learning acceleration logic, e.g., a fixed function matrix multiplication logic, in order to implement the stored processes and/or optimize the machine learning logic training and interface.

One type of algorithm suitable for use in machine learning modules as described herein is an artificial neural network or neural network, taking inspiration from biological neural

networks. An artificial neural network can, in a sense, learn to perform tasks by processing examples, without being programmed with any task-specific rules. A neural network generally includes connected units, neurons, or nodes (e.g., connected by synapses) and may allow for the machine learning program to improve performance. A neural network may define a network of functions, which have a graphical relationship. As an example, a feedforward network may be utilized, e.g., an acyclic graph with nodes arranged in layers.

A feedforward network (see, e.g., feedforward network 260 referenced in FIG. 2A) may include a topography with a hidden layer 264 between an input layer 262 and an output layer 266. The input layer 262, having nodes commonly referenced in FIG. 2A as input nodes 272 for convenience, communicates input data, variables, matrices, or the like to the hidden layer 264, having nodes 274. The hidden layer 264 generates a representation and/or transformation of the input data into a form that is suitable for generating output data. Adjacent layers of the topography are connected at the edges of the nodes of the respective layers, but nodes within a layer typically are not separated by an edge. In at least one embodiment of such a feedforward network, data is communicated to the nodes 272 of the input layer, which then communicates the data to the hidden layer 264. The hidden layer 264 may be configured to determine the state of the nodes in the respective layers and assign weight coefficients or parameters of the nodes based on the edges separating each of the layers, e.g., an activation function implemented between the input data communicated from the input layer 262 and the output data communicated to the nodes 276 of the output layer 266. It should be appreciated that the form of the output from the neural network may generally depend on the type of model represented by the algorithm. Although the feedforward network 260 of FIG. 2A expressly includes a single hidden layer 264, other embodiments of feedforward networks within the scope of the descriptions can include any number of hidden layers. The hidden layers are intermediate the input and output layers and are generally where all or most of the computation is done.

Neural networks may perform a supervised learning process where known inputs and known outputs are utilized to categorize, classify, or predict a quality of a future input. However, additional or alternative embodiments of the machine learning program may be trained utilizing unsupervised or semi-supervised training, where none of the outputs or some of the outputs are unknown, respectively. Typically, a machine learning algorithm is trained (e.g., utilizing a training data set) prior to modeling the problem with which the algorithm is associated. Supervised training of the neural network may include choosing a network topology suitable for the problem being modeled by the network and providing a set of training data representative of the problem. Generally, the machine learning algorithm may adjust the weight coefficients until any error in the output data generated by the algorithm is less than a predetermined, acceptable level. For instance, the training process may include comparing the generated output produced by the network in response to the training data with a desired or correct output. An associated error amount may then be determined for the generated output data, such as for each output data point generated in the output layer. The associated error amount may be communicated back through the system as an error signal, where the weight coefficients assigned in the hidden layer are adjusted based on the error signal. For instance, the associated error amount (e.g., a value between -1 and 1) may be used to modify the previous coefficient, e.g., a propagated value. The machine learning

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algorithm may be considered sufficiently trained when the associated error amount for the output data is less than the predetermined, acceptable level (e.g., each data point within the output layer includes an error amount less than the predetermined, acceptable level). Thus, the parameters

5 determined from the training process can be utilized with new input data to categorize, classify, and/or predict other values based on the new input data.

An additional or alternative type of neural network suitable for use in the machine learning program and/or module is a Convolutional Neural Network (CNN). A CNN is a type of feedforward neural network that may be utilized to model data associated with input data having a grid-like topology. In some embodiments, at least one layer of a CNN may include a sparsely connected layer, in which each output of a first hidden layer does not interact with each input of the next hidden layer. For example, the output of the convolution in the first hidden layer may be an input of the next hidden layer, rather than a respective state of each node of the first layer. CNNs are typically trained for pattern recognition, such as speech processing, language processing, and visual processing. As such, CNNs may be particularly useful for implementing optical and pattern recognition programs required from the machine learning program. A CNN includes an input layer, a hidden layer, and an output layer, typical of feedforward networks, but the nodes of a CNN input layer are generally organized into a set of categories via feature detectors and based on the receptive fields of the sensor, retina, input layer, etc. Each filter may then output data from its respective nodes to corresponding nodes of a subsequent layer of the network. A CNN may be configured to apply the convolution mathematical operation to the respective nodes of each filter and communicate the same to the corresponding node of the next subsequent layer. As an example, the input to the convolution layer may be a multidimensional array of data. The convolution layer, or hidden layer, may be a multidimensional array of parameters determined while training the model.

An exemplary convolutional neural network CNN is depicted and referenced as **280** in FIG. 2B. As in the basic feedforward network **260** of FIG. 2A, the illustrated example of FIG. 2B has an input layer **282** and an output layer **286**. However where a single hidden layer **264** is represented in FIG. 2A, multiple consecutive hidden layers **284A**, **284B**, and **284C** are represented in FIG. 2B. The edge neurons represented by white-filled arrows highlight that hidden layer nodes can be connected locally, such that not all nodes of succeeding layers are connected by neurons. FIG. 2C, representing a portion of the convolutional neural network **280** of FIG. 2B, specifically portions of the input layer **282** and the first hidden layer **284A**, illustrates that connections can be weighted. In the illustrated example, labels **W1** and **W2** refer to respective assigned weights for the referenced connections. Two hidden nodes **283** and **285** share the same set of weights **W1** and **W2** when connecting to two local patches.

Weight defines the impact a node in any given layer has on computations by a connected node in the next layer. FIG. 3 represents a particular node **300** in a hidden layer. The node **300** is connected to several nodes in the previous layer representing inputs to the node **300**. The input nodes **301**, **302**, **303** and **304** are each assigned a respective weight **W01**, **W02**, **W03**, and **W04** in the computation at the node **300**, which in this example is a weighted sum.

An additional or alternative type of feedforward neural network suitable for use in the machine learning program and/or module is a Recurrent Neural Network (RNN). An

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RNN may allow for analysis of sequences of inputs rather than only considering the current input data set. RNNs typically include feedback loops/connections between layers of the topography, thus allowing parameter data to be communicated between different parts of the neural network. RNNs typically have an architecture including cycles, where past values of a parameter influence the current calculation of the parameter, e.g., at least a portion of the output data from the RNN may be used as feedback/input in calculating subsequent output data. In some embodiments, the machine learning module may include an RNN configured for language processing, e.g., an RNN configured to perform statistical language modeling to predict the next word in a string based on the previous words. The RNN(s) of the machine learning program may include a feedback system suitable to provide the connection(s) between subsequent and previous layers of the network.

An example for a Recurrent Neural Network RNN is referenced as **400** in FIG. 4. As in the basic feedforward network **260** of FIG. 2A, the illustrated example of FIG. 4 has an input layer **410** (with nodes **412**) and an output layer **440** (with nodes **442**). However, where a single hidden layer **264** is represented in FIG. 2A, multiple consecutive hidden layers **420** and **430** are represented in FIG. 4 (with nodes **422** and nodes **432**, respectively). As shown, the RNN **400** includes a feedback connector **404** configured to communicate parameter data from at least one node **432** from the second hidden layer **430** to at least one node **422** of the first hidden layer **420**. It should be appreciated that two or more and up to all of the nodes of a subsequent layer may provide or communicate a parameter or other data to a previous layer of the RNN network **400**. Moreover and in some embodiments, the RNN **400** may include multiple feedback connectors **404** (e.g., connectors **404** suitable to communicatively couple pairs of nodes and/or connector systems **404** configured to provide communication between three or more nodes). Additionally or alternatively, the feedback connector **404** may communicatively couple two or more nodes having at least one hidden layer between them, i.e., nodes of nonsequential layers of the RNN **400**.

In an additional or alternative embodiment, the machine learning program may include one or more support vector machines. A support vector machine may be configured to determine a category to which input data belongs. For example, the machine learning program may be configured to define a margin using a combination of two or more of the input variables and/or data points as support vectors to maximize the determined margin. Such a margin may generally correspond to a distance between the closest vectors that are classified differently. The machine learning program may be configured to utilize a plurality of support vector machines to perform a single classification. For example, the machine learning program may determine the category to which input data belongs using a first support vector determined from first and second data points/variables, and the machine learning program may independently categorize the input data using a second support vector determined from third and fourth data points/variables. The support vector machine(s) may be trained similarly to the training of neural networks, e.g., by providing a known input vector (including values for the input variables) and a known output classification. The support vector machine is trained by selecting the support vectors and/or a portion of the input vectors that maximize the determined margin.

As depicted, and in some embodiments, the machine learning program may include a neural network topography having more than one hidden layer. In such embodiments,

one or more of the hidden layers may have a different number of nodes and/or the connections defined between layers. In some embodiments, each hidden layer may be configured to perform a different function. As an example, a first layer of the neural network may be configured to reduce a dimensionality of the input data, and a second layer of the neural network may be configured to perform statistical programs on the data communicated from the first layer. In various embodiments, each node of the previous layer of the network may be connected to an associated node of the subsequent layer (dense layers). Generally, the neural network(s) of the machine learning program may include a relatively large number of layers, e.g., three or more layers, and are referred to as deep neural networks. For example, the node of each hidden layer of a neural network may be associated with an activation function utilized by the machine learning program to generate an output received by a corresponding node in the subsequent layer. The last hidden layer of the neural network communicates a data set (e.g., the result of data processed within the respective layer) to the output layer. Deep neural networks may require more computational time and power to train, but the additional hidden layers provide multistep pattern recognition capability and/or reduced output error relative to simple or shallow machine learning architectures (e.g., including only one or two hidden layers).

Referring now to FIG. 5 and some embodiments, an AI program 502 may include a front-end algorithm 504 and a back-end algorithm 506. The artificial intelligence program 502 may be implemented on an AI processor 520, such as the processing device 120, the processing device 220, and/or a dedicated processing device. The instructions associated with the front-end algorithm 504 and the back-end algorithm 506 may be stored in an associated memory device and/or storage device of the system (e.g., memory device 124 and/or memory device 224) communicatively coupled to the AI processor 520, as shown. Additionally or alternatively, the system may include one or more memory devices and/or storage devices (represented by memory 524 in FIG. 5) for processing use and/or including one or more instructions necessary for operation of the AI program 502. In some embodiments, the AI program 502 may include a deep neural network (e.g., a front-end network 504 configured to perform pre-processing, such as feature recognition, and a back-end network 506 configured to perform an operation on the data set communicated directly or indirectly to the back-end network 506). For instance, the front-end program 506 can include at least one CNN 508 communicatively coupled to send output data to the back-end network 506.

Additionally or alternatively, the front-end program 504 can include one or more AI algorithms 510, 512 (e.g., statistical models or machine learning programs such as decision tree learning, associate rule learning, recurrent artificial neural networks, support vector machines, and the like). In various embodiments, the front-end program 504 may be configured to include built in training and inference logic or suitable software to train the neural network prior to use (e.g., machine learning logic including, but not limited to, image recognition, mapping and localization, autonomous navigation, speech synthesis, document imaging, or language translation). For example, a CNN 508 and/or AI algorithm 510 may be used for image recognition, input categorization, and/or support vector training. In some embodiments and within the front-end program 504, an output from an AI algorithm 510 may be communicated to a CNN 508 or 509, which processes the data before communicating an output from the CNN 508, 509 and/or the

front-end program 504 to the back-end program 506. In various embodiments, the back-end network 506 may be configured to implement input and/or model classification, speech recognition, translation, and the like. For instance, the back-end network 506 may include one or more CNNs (e.g., CNN 514) or dense networks (e.g., dense networks 516), as described herein.

For instance and in some embodiments of the AI program 502, the program may be configured to perform unsupervised learning, in which the machine learning program performs the training process using unlabeled data, e.g., without known output data with which to compare. During such unsupervised learning, the neural network may be configured to generate groupings of the input data and/or determine how individual input data points are related to the complete input data set (e.g., via the front-end program 504). For example, unsupervised training may be used to configure a neural network to generate a self-organizing map, reduce the dimensionality of the input data set, and/or to perform outlier/anomaly determinations to identify data points in the data set that falls outside the normal pattern of the data. In some embodiments, the AI program 502 may be trained using a semi-supervised learning process in which some but not all of the output data is known, e.g., a mix of labeled and unlabeled data having the same distribution.

In some embodiments, the AI program 502 may be accelerated via a machine learning framework 520 (e.g., hardware). The machine learning framework may include an index of basic operations, subroutines, and the like (primitives) typically implemented by AI and/or machine learning algorithms. Thus, the AI program 502 may be configured to utilize the primitives of the framework 520 to perform some or all of the calculations required by the AI program 502. Primitives suitable for inclusion in the machine learning framework 520 include operations associated with training a convolutional neural network (e.g., pools), tensor convolutions, activation functions, basic algebraic subroutines and programs (e.g., matrix operations, vector operations), numerical method subroutines and programs, and the like.

It should be appreciated that the machine learning program may include variations, adaptations, and alternatives suitable to perform the operations necessary for the system, and the present disclosure is equally applicable to such suitably configured machine learning and/or artificial intelligence programs, modules, etc. For instance, the machine learning program may include one or more long short-term memory (LSTM) RNNs, convolutional deep belief networks, deep belief networks DBNs, and the like. DBNs, for instance, may be utilized to pre-train the weighted characteristics and/or parameters using an unsupervised learning process. Further, the machine learning module may include one or more other machine learning tools (e.g., Logistic Regression (LR), Naive-Bayes, Random Forest (RF), matrix factorization, and support vector machines) in addition to, or as an alternative to, one or more neural networks, as described herein.

FIG. 6 is a flow chart representing a method 600, according to at least one embodiment, of model development and deployment by machine learning. The method 600 represents at least one example of a machine learning workflow in which steps are implemented in a machine learning project.

In step 602, a user authorizes, requests, manages, or initiates the machine-learning workflow. This may represent a user such as human agent, or customer, requesting machine-learning assistance or AI functionality to simulate intelligent behavior (such as a virtual agent) or other

machine-assisted or computerized tasks that may, for example, entail visual perception, speech recognition, decision-making, translation, forecasting, predictive modelling, and/or suggestions as non-limiting examples. In a first iteration from the user perspective, step 602 can represent a starting point. However, with regard to continuing or improving an ongoing machine learning workflow, step 602 can represent an opportunity for further user input or oversight via a feedback loop.

In step 604, data is received, collected, accessed, or otherwise acquired and entered as can be termed data ingestion. In step 606 the data ingested in step 604 is pre-processed, for example, by cleaning, and/or transformation such as into a format that the following components can digest. The incoming data may be versioned to connect a data snapshot with the particularly resulting trained model. As newly trained models are tied to a set of versioned data, preprocessing steps are tied to the developed model. If new data is subsequently collected and entered, a new model will be generated. If the preprocessing step 606 is updated with newly ingested data, an updated model will be generated. Step 606 can include data validation, which focuses on confirming that the statistics of the ingested data are as expected, such as that data values are within expected numerical ranges, that data sets are within any expected or required categories, and that data comply with any needed distributions such as within those categories. Step 606 can proceed to step 608 to automatically alert the initiating user, other human or virtual agents, and/or other systems, if any anomalies are detected in the data, thereby pausing or terminating the process flow until corrective action is taken.

In step 610, training test data such as a target variable value is inserted into an iterative training and testing loop. In step 612, model training, a core step of the machine learning work flow, is implemented. A model architecture is trained in the iterative training and testing loop. For example, features in the training test data are used to train the model based on weights and iterative calculations in which the target variable may be incorrectly predicted in an early iteration as determined by comparison in step 614, where the model is tested. Subsequent iterations of the model training, in step 612, may be conducted with updated weights in the calculations.

When compliance and/or success in the model testing in step 614 is achieved, process flow proceeds to step 616, where model deployment is triggered. The model may be utilized in AI functions and programming, for example to simulate intelligent behavior, to perform machine-assisted or computerized tasks, of which visual perception, speech recognition, decision-making, translation, forecasting, predictive modelling, and/or automated suggestion generation serve as non-limiting examples.

FIG. 7 is a representative flow chart illustrating the steps for training an AI engine for real-time monitoring of false positives during user registration. At block 705, the system receives a first set of data from at least one AI security model. In some embodiments, the first set of data is received from one AI security model. Additionally, or alternatively, the first set of data is received from more than one AI security model. The data received from the at least one security model includes input data into the at least one AI security model, output data from the at least one AI security model. Input data includes data indicative of user registration. In some embodiments, data indicative of user registration includes personally identifiable information such as full name, personally identifiable information such as social security number or tax identification number. In some

embodiments, output data from the at least one AI security model includes a classification of the input data. In some embodiments, the classification is a negative or positive for fraudulent behavior.

At block 710, the system receives a second set of data from a false positive database. In some embodiments the false positive database comprises information related to users who have reported false positives with the at least one AI security model. At block 715, the system correlates the first set of data with the second set of data. Correlating the two sets of data includes coupling the false positive from the false positive database with the appropriate input and output data associated with the instance of the user registration that prompted the AI security model to identify the user registration as positive for fraud.

At block 720, the system creates a training dataset and a test dataset from the correlated data. In some embodiments, the data is split in a 70/30 manner, 70% of the data is added to the training dataset and 30% of the data is added to the test dataset. In other embodiments, the data is split in a 50/50 manner. In some embodiments, the data is split in a 60/40 manner. Additionally, or alternatively, the data may be split in a 30/70 manner or a 40/60 manner.

At block 725, the system will use the training dataset to train a false positive identification model to identify false positives. The identification false positives may include labelling or classifying the positive data as false. In some embodiments, the false positive identification model includes a neural network as described above in the detailed description of the present disclosure. Additionally, or alternatively, the false positive identification model may be a concurrent neural network (CNN). In some embodiments, the false positive identification model may be a recurrent neural network (RNN). Additionally, or alternatively, the false positive identification model may be a feed-forward network. The false positive identification model may, in some embodiments, include a Bayesian machine learning algorithm. In some embodiments, the Bayesian machine learning algorithm is a Naive Bayes algorithm.

Training the false positive identification model follows the steps illustrated in flow chart 600 of FIG. 6. This includes the steps represented at block 730, using the test dataset to evaluate the false positive identification model. The steps represented in block 725 and 730 may be repeated until desired evaluation results are received. Once desired evaluation results are received the system will proceed to block 732, where an AI engine will be applied to user registration. The AI engine may comprise an AI security model and the false positive identification model. In some embodiments, the AI security model includes a neural network as described above in the detailed description of the present disclosure. Additionally, or alternatively, the AI security model may be a concurrent neural network (CNN). In some embodiments, the AI security model may be a recurrent neural network (RNN). Additionally, or alternatively, the AI security model may be a feed-forward network. The AI security model may, in some embodiments, include a Bayesian machine learning algorithm. In some embodiments, the Bayesian machine learning algorithm is a Naive Bayes algorithm. In some embodiments the AI security model includes more than one algorithm, or machine learning method.

When applying the AI engine, the system may follow the steps represented in flow chart 800 shown in FIG. 8. At block 805, the system may receive user information from a user attempting to register a user account. Upon receiving the user information, the system may, at block 807, extract

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data from the user information. The extracted data may include personally identifiable information. In some embodiments, personally identifiable information includes a social security number or a tax identification number. In some embodiments, personally identifiable information may include a user name. Additionally, or alternatively, personally identifiable information may include an address. In some embodiments, personally identifiable information may include a phone number. Additionally, or alternatively, personally identifiable information may include an email address. In some embodiments the extracted data includes an IP address from where the user is attempting to register the new account. Additionally, or alternatively, the extracted data may include a time and location from where the user is attempting to register the new account.

At block 810 the system will apply the AI security model, as described above. In some embodiments the AI security model applied in this step may be the same AI security model used to train the false positive identification model described in the flow chart 700. Additionally, or alternatively, the AI security model is a different AI security model used to train the false positive identification model described in flow chart 700.

After applying the at least one AI security model, the system will receive a result from the AI security model. The result will either be a positive result (fraudulent) or a negative result (not fraudulent). If the system receives a negative result from the AI security model, at block 820, then the system will allow the user registration, block 845. However, if the system receives a positive result from the AI security model, block 815, the system will apply the false positive identification model, block 825. The false positive identification model will provide the system a result, the result from the false positive identification model may be either true (this was a true positive result from the AI security model), or a false (this was a false positive result from the AI security model). Upon receiving a false result from the false positive identification model, block 830, the system will allow the user registration, block 845. If the system receives a true result from the false positive identification model, block 835, the system will block the user registration.

In some embodiments, applying the AI engine further includes periodically testing the model by determining a false positive rate. Based on the false positive rate, the system may trigger an alert to modify the AI security model. In some embodiments, modifying the AI security model includes retraining the AI security model. Retraining the AI security model includes the steps represented in flow chart 600. In other embodiments, modifying the AI security model includes selecting a new AI security model. In other embodiments, modifying the AI security model may include both retraining the AI security model and selecting a new AI security model. In some embodiments, when the AI security model contains more than one algorithm or model, only one of the algorithms or models may be modified. Additionally, or alternatively, when the AI security model contains more than one algorithm or model, more than one of the algorithms or models may be modified. In some embodiments, when the AI security model contains more than one algorithm or model, all of the algorithms or models may be modified.

Particular embodiments and features have been described with reference to the drawings. It is to be understood that these descriptions are not limited to any single embodiment or any particular set of features. Similar embodiments and features may arise or modifications and additions may be

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made without departing from the scope of these descriptions and the spirit of the appended claims.

What is claimed is:

1. A system for training an AI engine for real-time monitoring of false positives during user registration comprising:

- at least one processor;
- a communication interface communicatively coupled to the at least one processor; and
- a memory device storing executable code that, when executed, causes the processor to:
 - receive a first set of data from at least one AI security model, wherein the data comprises input data from the at least one AI security model and output data from the at least one AI security model, wherein the input data comprises data indicative of user registration, wherein the output data comprises a classification;
 - receive a second set of data from a false positive database, wherein the false positive database comprises a list of false positive instances;
 - correlate the first set of data with the second set of data to create a correlated dataset;
 - from the correlated dataset, create a training dataset and a test dataset;
 - use the training dataset to train a false positive identification model to identify false positives;
 - use the test dataset to perform an evaluation of the false positive identification models ability to identify false positives; and
 - based on the results of the evaluation, apply an AI engine to a user registration system, wherein the AI engine comprises an AI security model and the false positive identification model;

wherein the at least one AI security model is configured to compare user information to personally identifiable information from one or more public or private databases.

2. The system for training an AI engine according to claim 1, wherein applying the AI engine comprises:

- receive the user information from a user attempting to register a new account;
- upon receiving user information, extract data from the user information;
- apply the at least one AI security model to the extracted data;
- receive a result from the at least one AI security model, wherein the result is either a negative result or a positive result;
- upon receiving the negative result from the AI security model, allow user registration;
- upon receiving the positive result from the AI security model, apply the false positive identification model;
- receive a result from the false positive identification model, wherein the result may be a false result or a true result;
- upon receiving the false result from the false positive identification model, allow user registration; and
- upon receiving the true result from the false positive identification model, block user registration.

3. The system for training an AI engine according to claim 2, wherein applying the AI engine further comprises

- determining a false positive rate,
- based on the false positive rate, triggering an alert to modify the at least one AI security model.

4. The system for training an AI engine according to claim 3, wherein modifying the AI security model comprises:

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retraining the AI security model, or
selecting a new AI security model.

5. The system for training an AI engine according to claim 1, wherein the false positive identification model comprises a neural network.

6. The system for training an AI engine according to claim 5, wherein the neural network is selected from the group consisting of a convolution neural network (CNN), a recurrent neural network (RNN), and a feed-forward network.

7. The system for training an AI engine according to claim 1, wherein the false positive identification model comprises a Bayesian machine learning algorithm.

8. A method for training an AI engine for real-time monitoring of false positives during user registration, the method comprising:

receiving a first set of data from at least one AI security model, wherein the data comprises input data from the at least one AI security model and output data from the at least one AI security model, wherein the input data comprises data indicative of user registration, wherein the output data comprises a classification,

receiving a second set of data from a false positive database, wherein the false positive database comprises a list of false positive instances;

correlating the first set of data with the second set of data to create a correlated dataset;

creating a training dataset and a test dataset from the correlated dataset;

training a false positive identification model to identify false positives using the training dataset;

evaluating the false positive identification model using the test dataset;

altering the false positive identification model, based on results of the evaluation; and

applying an AI engine to user registration, based on the results of the evaluation, wherein the AI engine comprises an AI security model and the false positive identification model;

wherein the at least one AI security model is configured to compare user information to personally identifiable information from one or more public or private databases.

9. The method for training an AI engine according to claim 8, wherein applying the false positive identification model comprises:

receiving the user information;

extracting content from user information;

applying the AI security model to the extracted data;

receiving a result from the AI security model, wherein the result is either a positive result or a negative result;

allowing user registration, based on receiving the negative result;

applying the false positive identification model, based on receiving the positive result;

receiving a result from the false positive identification model, wherein the result is either a false result or a true result;

allowing user registration, based on receiving the false result; and

blocking user registration, based on receiving a true result.

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10. The method for training an AI engine according to claim 9, wherein applying the false positive identification model further comprises:

determining a false positive rate, and

triggering an alert to modify the AI security model.

11. The method for training an AI engine according to claim 8, wherein the false positive identification model comprises a neural network.

12. The method for training an AI engine according to claim 11, wherein the neural network is selected from the group consisting of a convolution neural network (CNN), a recurrent neural network (RNN), and a feed-forward network.

13. The method for training an AI engine according to claim 8, wherein the false positive identification model comprises a Bayesian machine learning algorithm.

14. A system for eliminating false positives comprising:
at least one processor;

a communication interface communicatively coupled to the at least one processor; and

a memory device storing executable code that, when executed, causes the processor to:

receive user information from a user attempting to register a account;

apply an AI security model to the user information, wherein the security model is configured to determine whether or not the user is attempting fraud;

receive a result from the AI security model, wherein the result is either a positive result or a negative result;

apply a false positive identification model, based on receiving the positive result;

receive a result from the false positive identification model, wherein the result is either a false result or a true result; and

allow user registration, based on receiving the false result;

wherein the at least one AI security model is configured to compare user information to personally identifiable information from one or more public or private databases.

15. The system for eliminating false positives according to claim 14, wherein the memory device further comprises executable code that, when executed, causes the processor to:

determine a false positive rate; and

based on the false positive rate, trigger an alert to modify the AI security model.

16. The system for eliminating false positives according to claim 15, wherein modifying the AI security model comprises:

retraining the AI security model, or

selecting a new AI security model.

17. The system for eliminating false positives according to claim 14, wherein the false positive identification model comprises a neural network comprising a convolution neural network (CNN), a recurrent neural network (RNN), and a feed-forward network.

18. The system for eliminating false positives according to claim 14, wherein the false positive identification model comprises a Bayesian machine learning algorithm.

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